



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

**CANDIDATE
NAME**

CLASS

CHEMISTRY

Preliminary Examination
 Paper 2 Structured Questions (SPA)

9647/02
20 August 2015
2 hours

Candidates answer on the Question Paper.
 Additional Materials: Data Booklet.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
 Write in dark blue or black pen on both sides of the paper.
 You may use a soft pencil for any diagrams, graphs or rough work.

Answer **all** questions in the space provided.
 You are advised to spend **30 minutes** on question 1.

At the end of the examination, fasten all your work securely together.
 The number of marks is given in the brackets [] at the end of each question or part questions.

For Examiner's Use	
P1	/40
P2	/72
P3	/80
GRAND TOTAL	/192
%	
GRADE	

For Examiner's Use	
1(P)	/12
2	/25
3	/15
4	/20
TOTAL P2	/72

Answer all questions

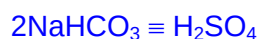
1	The Solvay process is a major industrial method for the production of soda ash. As the ingredients for this process are readily available and inexpensive, Solvay-based chemical plants produce almost three-quarters of the supply worldwide. In the first step of the manufacturing process, sodium bicarbonate, NaHCO_3, is initially yielded. The reaction involves carbon dioxide gas passing through concentrated aqueous brine solution and ammonia gas: $\text{NaCl (aq)} + \text{CO}_2 \text{ (g)} + \text{NH}_3 \text{ (g)} + \text{H}_2\text{O (l)} \rightarrow \text{NaHCO}_3 \text{ (s)} + \text{NH}_4\text{Cl (l)} \text{ --- (1)}$ Solid sodium bicarbonate is then precipitated out and filtered from the hot ammonium chloride solution. Quality control tests on sodium bicarbonate sample X found trace amounts of contaminants including by-products and fragments from the alloy metal pipelines.
(a)	You are required to design an experiment to determine the percentage purity of sample X. In your plan, you will collect and measure the volume of carbon dioxide gas produced from the reaction between a known quantity of baking soda and sulfuric acid: $2\text{NaHCO}_3 \text{ (s)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{Na}_2\text{SO}_4 \text{ (aq)} + 2 \text{CO}_2 \text{ (g)} + 2 \text{H}_2\text{O (l)} \text{ --- (2)}$ You are provided with: • Sample X • $0.200 \text{ mol dm}^{-3}$ sulfuric acid • All common apparatus normally found in a school laboratory In your plan you should: • include clear calculations on the mass and volume of the reactants to be used in your experiment, stating any assumptions • draw a fully-labelled diagram of the assembled apparatus you would use • give a full description of the essential experimental procedures [6]

- Assume r.t.p
- Assume volume of carbon dioxide collected to be 80 cm^3 (accept 50 cm^3 to 100 cm^3 , which are between $\frac{1}{2}$ capacity and max capacity of syringe)

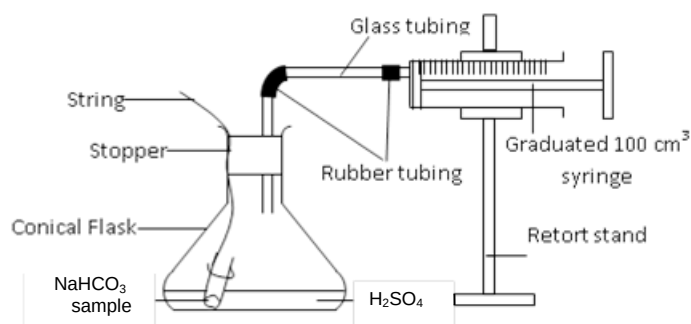
$$\begin{aligned}\text{Amt of CO}_2 &= 80/24000 \text{ mol} \\ &= 3.33 \times 10^{-3} \text{ mol}\end{aligned}$$



$$\begin{aligned}\text{Mass of NaHCO}_3 &= 3.33 \times 10^{-3} \text{ mol} \times 84 \text{ g} \\ &= \underline{\underline{0.28\text{g}}}\end{aligned}$$



$$\begin{aligned}\text{Minimum volume of H}_2\text{SO}_4 &= [(3.33 \times 10^{-3})/2] / 0.20 \\ &= \underline{\underline{8.33 \text{ cm}^3}}\end{aligned}$$



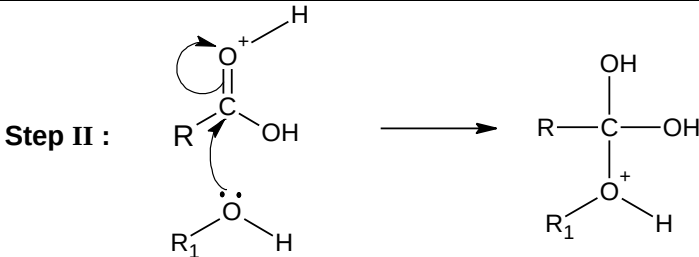
1. Using an **electronic balance/ weighing balance** (1), weigh (1) accurately about **0.28g to 0.30g sodium bicarbonate** (1) in a weighing bottle (1).
2. Using a **measuring cylinder** (1), transfer **10 cm³** (1) **sulfuric acid** (1) into a **250cm³ conical flask** (1).
3. Carefully **lower the weighing bottle** (1) containing the sodium bicarbonate into the conical flask, ensuring that the reactants do not mix.
4. Set up the apparatus above and **record the initial reading** (1) **x cm³** on the **gas syringe** (1).
5. **Shake the conical flask to topple** (1) the weighing bottle, and **swirl** (1) the conical flask thoroughly to mix the reactants.
6. When no more effervescence is observed, **record the final reading** (1) **y cm³** on the gas syringe. The **volume of CO₂ collected is (y-x) cm³** (1).

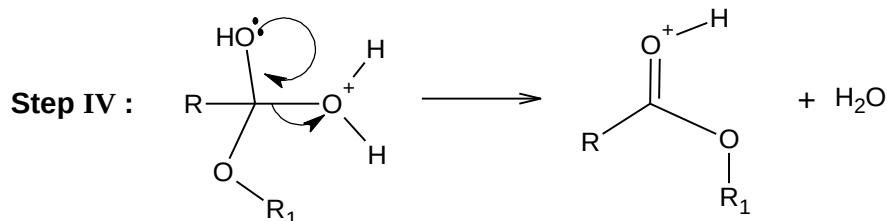
OR

1. Using an **electronic balance/ weighing balance** (1), weigh accurately about **0.28g to 0.30g sodium bicarbonate** (1) in a **conical flask** (1).
2. Fill a **10 cm³ of sulphuric acid** into **thistle funnel** / Fill **graduated dropping funnel** (1) with **sulfuric acid** (1).
3. Set up the apparatus above and **record the initial reading** (1) **x cm³** on the **gas syringe** (1).
4. **Open the tap** of the thistle funnel to **transfer** (1) all the sulfuric acid into the conical flask. / Take note of the initial reading of the **graduated dropping funnel**. Open the tap to add **10 cm³ of sulfuric acid** into the conical flask and close the tap. (1)
5. **Swirl** the conical flask thoroughly to mix the reactants. (1)
6. When no more effervescence is observed, record the final reading **y cm³** on the gas syringe. The volume of CO₂ collected is **(y-x) cm³**.

	(b)	State one precaution that should be taken to ensure the reliability of the experiment. [1]
		• <u>Grease the syringe</u> to allow for <u>smooth movement of plunger during gas collection</u> OR • <u>Wait until no more effervescence is observed</u> before taking final volume of gas to allow for <u>temperature and pressure to equilibrate with surroundings</u>.
	(c)	The contaminants were suspected to contain the cations Pb^{2+} and Zn^{2+} as well as the by-product NH_4Cl. Outline a series of test-tube experiments in order to identify the presence of these ions. Your plan should include all chemical tests, expected observations and deductions. Identification by elimination method is not allowed in your plan. Equations are not required in your plan [5]

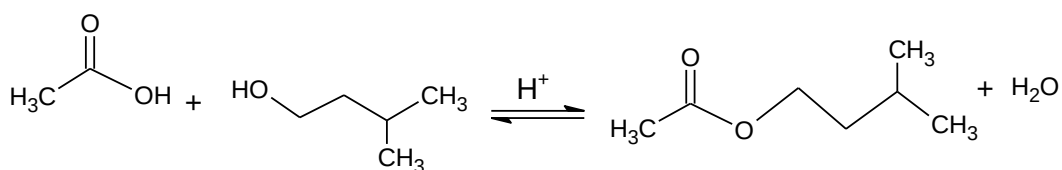
Steps	Test	Observations	Identification of ion
1	Add <u>1 cm³</u> of the solutions to four separate <u>test tubes</u> . Add a <u>few drops</u> of <u>aqueous silver nitrate</u> to each test tube.	<u>White ppt</u> of AgCl, and white ppt <u>dissolves to form colourless solution</u> <u>No ppt formed in other 3 test tubes</u>	<u>Cl⁻</u>
2	Add <u>1 cm³</u> of <u>the remaining solutions</u> to three separate <u>test tubes</u> . Add <u>NH₃ (aq)</u> <u>dropwise with shaking, until in excess</u> using a <u>dropper</u> to each test tube at <u>r.t.p.</u>	<u>White ppt</u> Pb(OH) ₂ and Zn(OH) ₂ will be produced in 2 test tubes, and <u>ppt is soluble in excess NH₃(aq) for 1 test tube, producing colourless solution</u> , [Zn(NH ₃) ₄] ²⁺ <u>White ppt</u> Pb(OH) ₂ remains insoluble The last test tube has no ppt	<u>Zn²⁺</u>
3	Add <u>1 cm³</u> of <u>the remaining solutions</u> to two separate <u>test tubes</u> . Add NaOH(aq) to each test tube and <u>gently warm the test tube</u> .	<u>Pungent gas which turns moist red litmus paper blue is produced</u>	<u>NH₄⁺</u>
4	Add <u>1 cm³</u> of <u>the last solution</u> to a <u>test tube</u> . Add <u>Potassium iodide (aq)</u>	<u>Yellow ppt</u> PbI ₂ is formed	<u>Pb²⁺</u>
[Total : 12]			

2	Esters are derivatives of carboxylic acids. Simple esters tend to have pleasant, fruity odors and are widely used as flavours and fragrances. Esterification generally refers to the formation of esters from alcohol and carboxylic acids, as shown in equation 1. This esterification mechanism is thought to involve 5 steps. $\text{RCOOH} + \text{R}'\text{OH} \rightleftharpoons \text{RCOOR}' + \text{H}_2\text{O} \quad \text{equation 1}$ Step I : The carbonyl carbon on the carboxylic acid gains a proton. $\text{RCOOH} + \text{H}^+ \rightleftharpoons \begin{array}{c} \text{O}^+ - \text{H} \\ \\ \text{R} - \text{C} \\ \\ \text{OH} \end{array}$ Step II : Next, the electron-rich oxygen atom of the alcohol attacks the carbon atom of the carboxylic acid to form a positively-charged tetrahedral intermediate. Step III : This is followed by transferring a proton to the hydroxyl oxygen of the carboxyl group. Step IV : Subsequently, water is removed to form the positively charged ester Step V : The ester is formed when a proton is transferred to the base.		
	(a)	(i)	Suggest why protonation in Step I is required before nucleophilic attack in Step II can occur. The protonation <u>activates the carbonyl carbon</u> toward nucleophilic attack. This is because O would withdraw greater electrons from C, making it even <u>more partial positive</u>.
		(ii)	Using the information given on page 5 to propose the mechanism for only steps II and IV in the given boxes below. In your answer, you should include all charges and lone pairs and show the movement of electrons by curly arrows. [3]
			Step II : 



- (b) Following the success of the movie, The Minions, food manufacturers have sought to include banana flavours in their products. The compound that gives characteristic banana flavour is isoamyl acetate.

Isoamyl acetate can be synthesised via esterification as follows :



ethanoic acid

3-methylbutan-1-ol

isoamyl acetate

Data about the compounds is given below.

Compound	Boiling point/ °C	Density/ g cm ⁻³	M _r	Solubility in water
3-methylbutan-1-ol	130	0.809	88.0	slightly soluble
ethanoic acid	118	1.05	60.0	soluble
concentrated sulfuric acid	330	1.84	98.0	soluble
water	100	1.00	18.0	-
isoamyl acetate	142	0.876	130.0	slightly soluble

Synthesis of crude isoamyl acetate

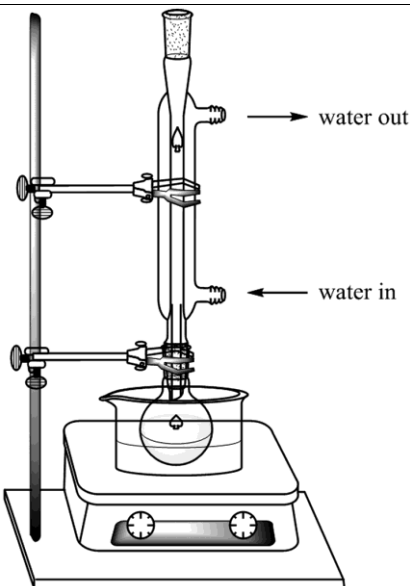
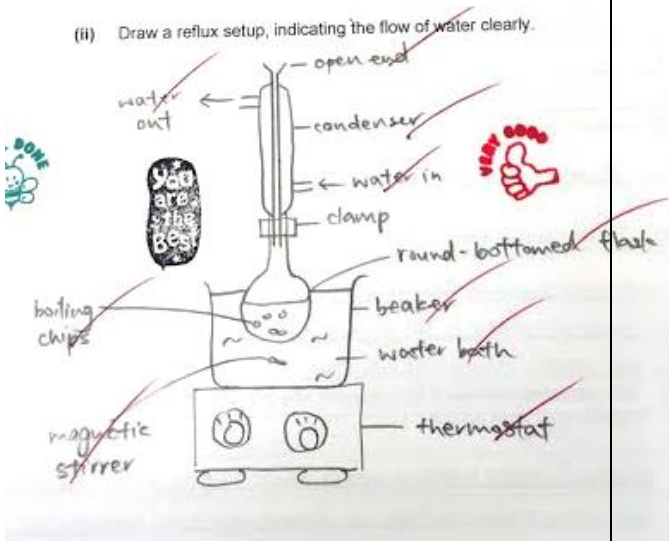
1. Add 3.2 g of 3-methylbutan-1-ol into a round bottom flask. Add 5 cm³ of ethanoic acid and 3 boiling chips into the flask.
2. Add 6 drops of concentrated sulfuric acid.
3. Construct a reflux set up and heat the mixture for 30 min.
4. At the end of the reflux period, remove the flask from the heat and let the mixture cool to room temperature.

- (i) Using the information given, determine, by calculation, which reagent is in excess.

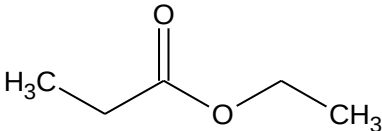
Amount of 3-methylbutan-1-ol = $3.2/88 = 0.03636 \text{ mol}$

Amount of ethanoic acid = $(5 \times 1.05)/60 = 0.0875 \text{ mol}$

Since the ratio of 3- methylbutan-1-ol : ethanoic acid = 1:1,
ethanoic acid is in excess.

		(ii)	Draw a reflux setup, indicating the flow of water clearly.	[3]
			  Marking points: • Round bottom flask • Heating mantle • Water bath • Condenser • Flow of water	
		(c)	<u>Purification of crude isoamyl acetate</u> 5. Shake the crude isoamyl acetate with 25 cm³ water in a separating funnel and separate the aqueous layer from isoamyl acetate. Reject the aqueous layer. 6. Return the isoamyl acetate to the funnel, cautiously add dilute aqueous sodium carbonate in the separating funnel, releasing the pressure at intervals.	
		(i)	Give two reasons why the addition of water in step 5 is necessary before step 6.	
			1. Remove <u>polar</u> impurities 2. Sodium carbonate <u>reacts</u> with the <u>acidic mixture</u> to give an <u>effervescence</u> of carbon dioxide. Adding water <u>dilutes the acid</u> to <u>prevent vigorous reaction</u> from occurring.	
		(ii)	In step 6, why is it important to release the pressure at intervals ?	

		<u>To prevent pressure build up</u> as carbon dioxide gas is formed.	
			[3]
(d)	<u>Purification of crude isoamyl acetate</u> 7. Transfer the isoamyl acetate into a conical flask and add some granular anhydrous sodium sulfate. Swirl the mixture until the liquid is clear. 8. Filter the isoamyl acetate into a clean, dry round-bottomed flask and distil it. Collect the fraction by boiling over a suitable range.		
	(i)	What is the role of anhydrous sodium sulfate?	
		<u>Dehydrating agent</u>	
	(ii)	Explain the significance of a clear liquid obtained after Step 7.	
		<u>Water</u> has been <u>removed</u> from the mixture by anhydrous sodium sulfate.	
	(iii)	With reference to Steps 5-8, suggest why the reactant you have determined in (b)(i) must be in excess.	
		It is easier to remove ethanoic acid from the crude product <u>using a carbonate</u> via <u>neutralisation</u> . OR Alcohol has a <u>relatively close boiling point</u> to isoamyl acetate, hence it is <u>more difficult to separate</u> in Step 8.	
			[3]
(e)	After separating isoamyl acetate from water, there will still be a very small amount of one of the original reactants present as an impurity. To remove this, the isoamyl acetate is shaken with reagent X. A reaction occurs with the impurity present in the isoamyl acetate, increasing its solubility.		
	(i)	Suggest the identity of the impurity and reagent X.	
		Impurity : 3-methylbutan-1-ol Reagent X :HCl (or any strong acid) OR sodium	
	(ii)	By using structural formula, write a chemical equation for this reaction.	
		If HCl is used: $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)_2 + \text{H}^+ \longrightarrow \text{H}-\text{O}^+-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)_2$ OR If Na is used: $2 \text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)_2 + 2\text{Na} \longrightarrow 2 \text{Na}^+\text{O}^--\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)_2 + \text{H}_2$	

		(iii)	Suggest why would adding reagent X increase the solubility of the impurity to aid its removal.
			The product formed is ionic and will be soluble in water because of <u>ion-dipole interactions</u> .
			[3]
	(f)		After the distillate was collected, it was discovered that the boiling point is 10°C higher than the expected temperature.
		(i)	Give a possible reason for the temperature difference.
			<u>Presence of impurities</u>
		(ii)	Outline another technique to obtain pure isoamyl acetate other than distillation.
			<u>Column chromatography</u> (but chromatography alone is enough) More polar substances will be more strongly adsorbed to the stationary phase and will be retained for a longer time. Less polar substances such as isoamyl acetate will be less strongly adsorbed and will be eluted (washed off) first. Watch this video for a better understanding. http://www.chem-ilp.net/labTechniques/ChromatographyVideo.htm
			[2]
	(g)		After purification, the mass of isoamyl acetate obtained is 2.77g. Calculate the percentage yield of this synthesis.
			<u>Expected mass of isoamyl acetate = 0.036364 x 130 = 4.727 g</u> <u>Percentage yield = (2.77 / 4.727) x 100% = 58.6%</u>
			[1]
	(h)		A bottle containing the product, isoamyl acetate, was wrongly labelled as ethyl propanoate. Describe a simple chemical test to distinguish between the two bottles, clearly stating the observation(s) made. Include balanced chemical equations for your chosen test.
			 Ethyl propanoate
			<u>Chemical Test :</u>

Add NaOH(aq), heat to hydrolyse the ester.
Add aq I₂ in aqueous NaOH heat to the hydrolysed products.

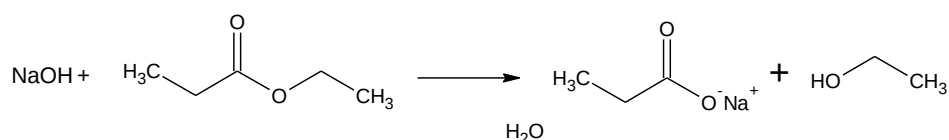
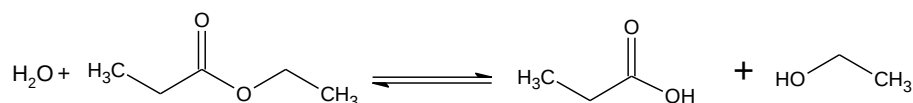
Observations :

Ethyl propylpropanoate will decolourise brown I₂ solution and a yellow ppt is observed.

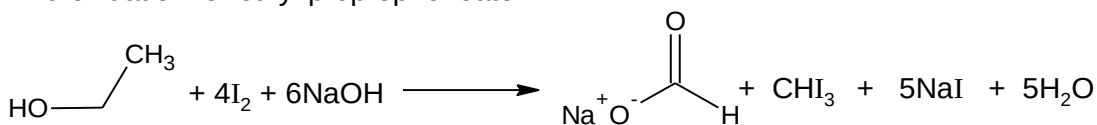
Isoamyl acetate remains brown and no yellow ppt is observed.

Equations :

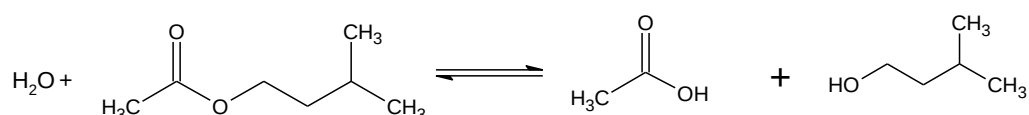
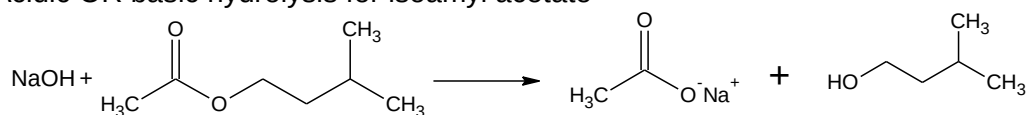
Acidic OR basic hydrolysis for ethyl propylpropanoate



Mild oxidation for ethyl propylpropanoate

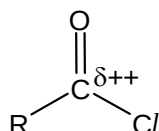


Acidic OR basic hydrolysis for isoamyl acetate



[5]

- (i) Using your chemical knowledge of the reactivity of acyl chlorides, suggest why ethanoyl chloride would be a better alternative to make isoamyl acetate.



The carbonyl C atom has **two** very electronegative atoms (O and Cl) bonded to it. Both O and Cl **exert electron-withdrawing effects** the carbonyl C atom has a **large positive charge (i.e. δ^{++})**. Therefore, more susceptible to nucleophilic substitution.

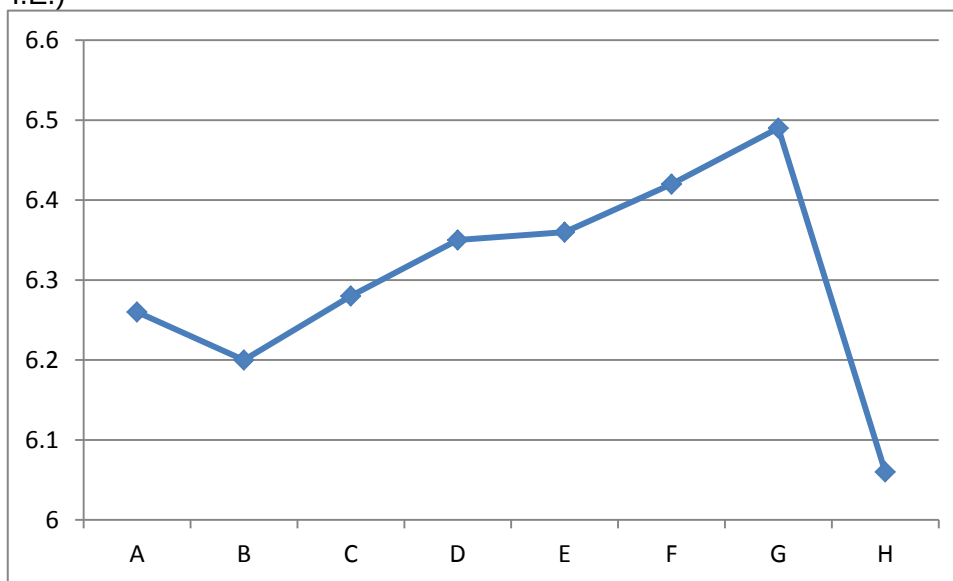
		AND A <u>higher yield</u> is obtained because the reaction with ethanoyl chloride is an <u>irreversible reaction</u>. OR It is <u>more economical</u> (or cheaper) because the reaction with ethanoyl chloride takes place at <u>room temperature</u>. OR <u>Faster</u> as reaction goes to completion.
		[2]
		[Total : 25]

- 3 The periodic table is a tabular arrangement of the chemical elements, organised by their atomic number, electronic configurations and recurring chemical properties. Dimitri Mendeleev is generally credited with the first publication of the first widely recognised Periodic Table in 1869.

Elements **A, B, C, D, E, F, G** and **H** are consecutive elements in the Periodic Table from periods 3 to 4.

The second ionisation energy (I.E.) of elements **A** to **H** are illustrated below.

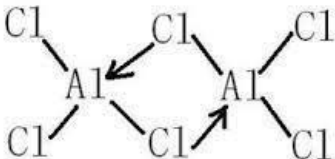
Log (2nd I.E.)

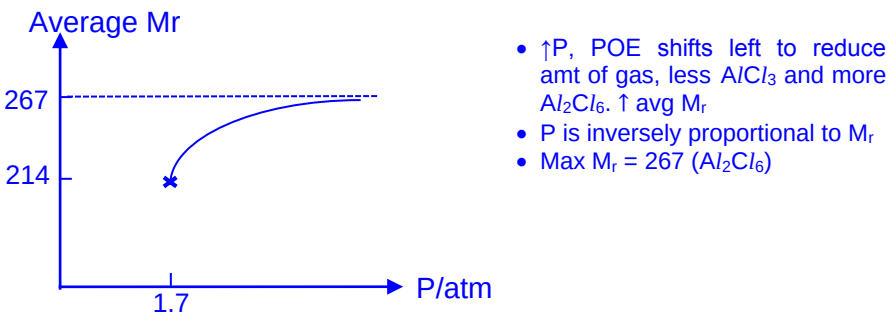


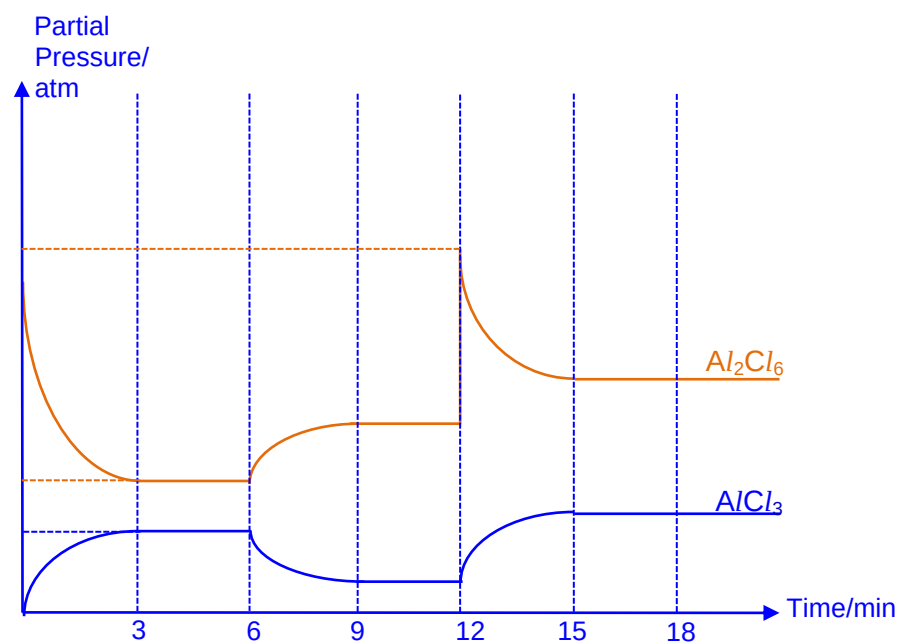
(a)	(i)	With the aid of a relevant equation, explain what is meant by the term second ionisation energy of the element D .
		$D^+(g) \rightarrow D^{2+}(g) + e$ Second ionisation energy of an element is the energy absorbed to remove one mole of electrons from one mole of singly charged positive gaseous ion to give one mole of doubly charged positive gaseous ions.
	(ii)	Account for the difference in second ionisation energy of A and B; D and E.
		[5] - The 3s electron to be removed from A⁺ is at a lower energy level compared to the 3p electrons removed from B⁺. More energy is required to overcome the stronger electrostatic forces of attraction between the 3s electron and the nucleus. Nuclear charge of E⁺ is greater than that of D⁺ while shielding effect is similar. Thus effective nuclear charge of E⁺ is greater than that of D⁺. Electrostatic forces of attraction between nucleus and electron to be removed is greater for E⁺ than that of D⁺.

		However, there is <u>interelectronic repulsion between the pair of electrons in the 3p orbital in E⁺</u> . Thus, the amount of energy required to remove the electron <u>increases by a small extent</u> .
	(b)	By interpreting the data above, deduce the identity of elements B and F and compare their melting points using structure and bonding. [2]
		B is silicon. Silicon has a <u>giant molecular/covalent structure</u>. <u>Larger</u> amount of energy is required to overcome the <u>strong covalent bonds between the Si atoms in a tetrahedral / extensive 3-D network</u>. Si has a high melting point. F is argon. Argon exist as simple <u>atoms</u>. <u>Lesser</u> amount of energy is required to overcome the <u>weaker van der Waal's forces of attraction between atoms</u>. Ar has a low melting point.
	(c)	When dissolved in water, the chlorides of elements B and C produce solutions that turn blue litmus red. Explain. [1]
		Electronegative chlorine causes partial positive charge on B and C. Water molecules attack the central atom donating lone pair of electrons into empty d-orbitals. <i>Cl</i> is substituted by OH⁻ liberating HCl. OR <u>Complete hydrolysis</u> takes place
	(d)	Oxides of A , B and H forms white solids. Suggest how they may be differentiated from one another. Write balanced chemical equations for any reactions that occur. [3]
		Oxides of A : Al₂O₃ Oxides of B : SiO₂ Oxides of H : CaO Firstly, <u>add NaOH</u> to the solids separately. Only Al₂O₃ will <u>dissolve</u>. $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4]$ Next, <u>add HCl</u> to the remaining solids separately. Only <u>CaO will dissolve</u>. $\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$ <u>SiO₂ remains as a solid.</u>
	(e)	(i) Describe what you would see when elements D is burned in air. Write equations for the reactions that occur.
		Observations: Sulfur burns with a <u>blue flame</u>, forming a colourless gas, SO₂. Equation: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$

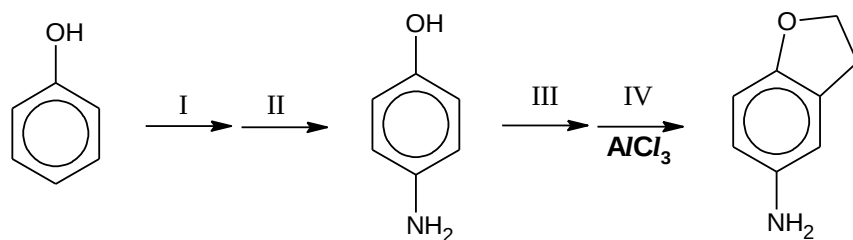
		(ii)	The oxides of D and G resulting from the reactions in (e)(i), both react with water. Write balanced equations for these two reactions and describe the colours observed when methyl orange is added to the resulting solutions. [4]
			$\text{K}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow 2 \text{KOH(aq)}$ <u>Yellow</u> for KOH $\text{SO}_2\text{(g)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{SO}_3\text{(aq)}$ <u>Red</u> for H_2SO_3
			[Total: 15]

4	Aluminium chloride is a white compound that is used in the product aluminium metal and in chemical industries. It exists as a dimer in liquid state and starts to vapourise at a temperature of 120°C. At even higher temperatures, the dimers dissociate into monomers as shown in the equation below. $\text{Al}_2\text{Cl}_6 \text{ (g)} \rightleftharpoons 2 \text{AlCl}_3 \text{ (g)}$		
	(a)	Draw the displayed formula of the vapour at 120°C, to show its bonding.	[1]
			
	(b)	2.63g of aluminium chloride was placed in a 300 cm ³ vessel and heated to 227 °C. The pressure inside the flask rose from 1.36 atm to 1.7 atm.	
	(i)	Calculate the average relative molecular mass of the mixture at 227 °C.	[1]
		$pV = \frac{m}{M_r} RT$ $(1.7 \times 10^5) (300 \times 10^{-6}) = \frac{2.63}{M_r} \times 8.31 \times 500$ average $M_r = 214.0$	
	(ii)	Hence, calculate the partial pressures of AlCl_3 and Al_2Cl_6 in the mixture.	[2]
		Let the mole fraction of Al_2Cl_6 be x , mole fraction of AlCl_3 will be $(1-x)$. $214 = 267(x) + 133.5(1-x)$ $x = 0.603$ $P_{\text{Al}_2\text{Cl}_6} = 0.603 \times 1.7 = 1.03 \text{ atm}$ $P_{\text{AlCl}_3} = (1-0.603) \times 1.7 = 0.675 \text{ atm or } 1.7-1.03 = 0.$	
	(iii)	Write an expression for K_p and calculate its value.	[2]
		$K_p = \frac{P_{\text{AlCl}_3}^2}{P_{\text{Al}_2\text{Cl}_6}}$ $= \frac{0.675^2}{1.025} = 0.442 \text{ atm}$	

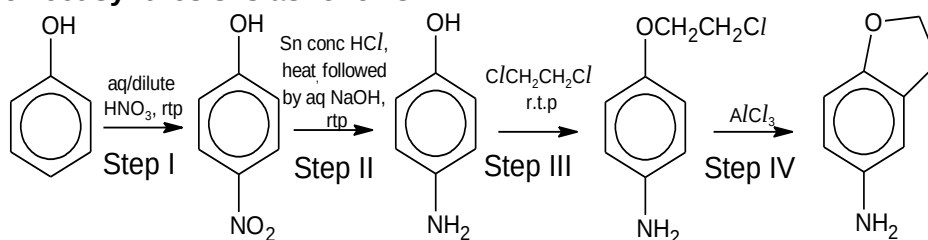
		(iv)	Use your answer in (b)(i) and (b)(iii) to sketch a graph to show how the average M_r of the mixture would vary with pressure starting from 1.7 atm.	[2]
				
		(v)	State and explain if the forward reaction is exothermic or endothermic.	[1]
			The forward reaction involves the <u>breaking of a Al-Cl bond</u> and thus is <u>endothermic</u> .	
		(vi)	Hence, illustrate in a pressure-time graph, the changes in partial pressures of Al_2Cl_6 and $AlCl_3$ when - the above equilibrium reaches equilibrium at the 3rd minute, - temperature decreases to 150 °C at the 6th minute. - partial pressure Al_2Cl_6 increases to 1.5 atm at the 12th minute. - addition of inert gas into the vessel at the 18th minute. You may assume for each change, the system will take about 3 minutes to re-establish the equilibrium again.	[4]



- (c) In the pharmaceutical industry, aluminium chloride is used as a catalyst in reactions such as Friedel-Crafts alkylation to attach an alkyl group onto a benzene ring. One synthesis of an organic compound involving $AlCl_3$ is shown below.



The correct synthesis is as follows : -

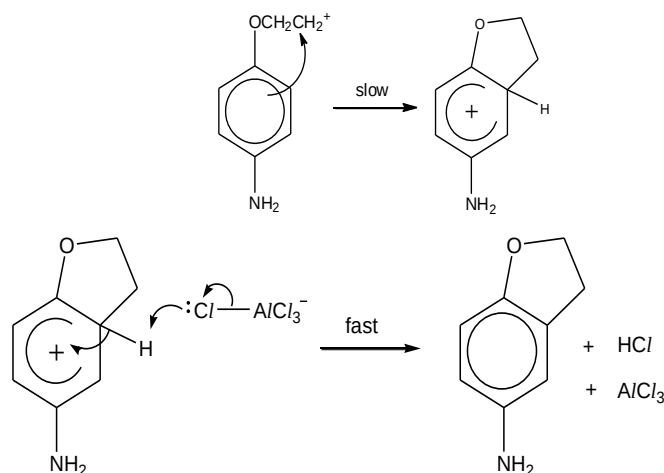


Step I : Phenols are activating, there is no need for a catalyst nor heat. (Easy) **electrophilic MONO-substitution** will take place in dilute solutions and room temperature.

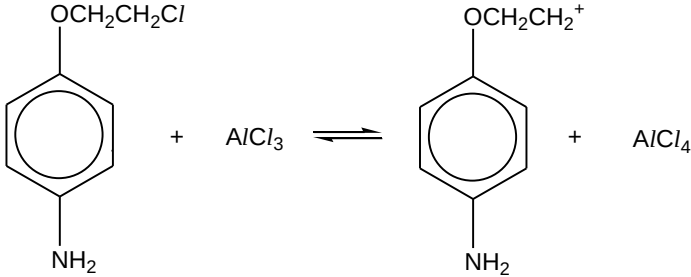
Step II : Sn and conc HCl undergoes **reduction** to form H_2 gas which reduces the $R-NO_2$ to $R-NH_3^+Cl^-$, an acidic amide salt. In order to regain the basic $R-NH_2$, **neutralisation** must be carried out with a strong base.

Step III : **Nucleophilic substitution** takes place when the reacting species, a nucleophile, $ClCH_2CH_2Cl$, reacts at the phenol where the first Cl atom leaves.

Step IV : The formation of the cyclic ring forms when **intra-electrophilic substitution** takes place and the next Cl leaves.



(i)	State the reagents and conditions for steps I and II.	[2]
	Step I : <u>Dilute HNO_3, rtp</u> Step II : <u>Sn in conc HCl/ heat, followed by NaOH (aq)</u>	
(ii)	State the type of reaction in steps III and IV.	[2]
	Step III : <u>Nucleophilic substitution</u> Step IV : <u>(Intra) electrophilic substitution</u>	
(iii)	State the reacting species used in step III.	[1]
	<u>$ClCH_2CH_2Cl$</u>	
(iv)	Explain why aluminium chloride can act as a Lewis acid in step IV.	[1]
	<u>Al is electron deficient/ has vacant energetically available orbitals to act as electron pair acceptor</u> , thus acting as a Lewis acid. (FYI)	

				
		(v)	Suggest why $AlCl_3$ would make a better Lewis acid than $AlBr_3$.	[1]
			Chlorine is <u>more electronegative</u> than bromine causing the aluminium in $AlCl_3$ to be <u>more electron deficient</u> than the aluminium in $AlBr_3$ making it better at accepting a pair of electrons from a halogen / Lewis acid.	
			[Total : 20 m]	

END OF PAPER 2